A Diagonal Counterexample to "Dimension Spectrum Encoded by D"

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Status

Claim status: candidate counterexample, likely valid. This packet addresses Chapter 5, Problem 5 of Eckstein–Iochum [1]. The problem asks whether the dimension spectrum of a spectral triple is always contained in the union of the pole sets of the D-only zeta functions $\zeta_{|D|^{-k},D}$. The construction below shows that this is false for regular spectral triples under the broad definitions used in the source.

Source Excerpt

The relevant question appears in Chapter 5, page 123 of the source PDF.

been accomplished only for a few examples listed on p. 20. Firstly, the existence of the dimension spectrum is by no means automatic — there exist spectral zeta functions admitting no meromorphic continuation (see [59] and [28, Section 5]). Secondly, even if we do have a meromorphic extension of ζ_D, the one of $\zeta_{T,D}$ does not come for free, even for $T \in \mathscr{A}$. In the worked out examples, one firstly unravels the meromorphic extension of the basic zeta function ζ_D and then constructs the ones for $\zeta_{T,D}$ by expressing the operators $T \in \Psi^0(\mathscr{A})$ in the eigenbasis of $\mathscr{D}$. Whereas the poles of ζ_D and $\zeta_{T,D}$ do not coincide in general (see Example 1.24), in all known cases we actually have $\mathrm{Sd} \subset \cup_{k\in\mathbb{N}} \mathfrak{P}(\zeta_{|D|^{-k},D})$. So it seems as if the whole dimension spectrum is actually encoded in the operator $\mathscr{D}$.
Is this a general fact or a specific property of the worked out examples?

6) *Heat traces and zeta functions.* In Chapter 3 we have spied into the intimate

Transcribed Open Question

The source observes that, in all worked-out examples known there,

$$\mathrm{Sd} \subset \bigcup_{k\in\mathbb{N}} \mathcal{P}(\zeta_{|D|^{-k},D}),$$

and then asks whether the whole dimension spectrum is actually encoded in the operator D, or whether this is only a property of the worked-out examples.

Proof Intuition

The spectrum of D controls the scalar zeta functions $\mathrm{Tr}\,|D|^{-s-k}$, but the algebra can insert bounded diagonal coefficients into the trace. If those coefficients are n^i, then the corresponding zeta function is $\sum n^{i-s} = \zeta(s-i)$, whose pole is shifted from 1 to $1+i$. Since the inserted diagonal unitary commutes with D, regularity is automatic and the operator is order zero. Thus the algebra, not just D, contributes a genuinely new pole.

Theorem 1. *There is a regular 1-dimensional spectral triple with simple dimension spectrum for which the minimal dimension spectrum is not contained in*

$$\bigcup_{k\in\mathbb{N}_0} \mathcal{P}(\zeta_{|D|^{-k},D}).$$

Proof. Let $\mathcal{H} = \ell^2(\mathbb{N})$ with standard basis $(e_n)_{n\geq 1}$, and define

$$De_n = ne_n.$$

Then D is positive self-adjoint with compact resolvent. Let U be the diagonal unitary

$$Ue_n = n^i e_n = e^{i\log n} e_n,$$

and let $\mathcal{A}$ be the concrete unital $*$-subalgebra of $\mathcal{B}(\mathcal{H})$ generated by U. Thus the representation of $\mathcal{A}$ on $\mathcal{H}$ is faithful by definition.

For every $a \in \mathcal{A}$, a is a bounded diagonal operator and $[D,a] = 0$. Hence $(\mathcal{A},\mathcal{H},D)$ is a spectral triple, and it is regular because all iterated commutators with $|D|$ vanish on $\mathcal{A}$ and on $[D,\mathcal{A}]$. Its metric dimension is 1, since

$$\operatorname{Tr}|D|^{-s} = \sum_{n=1}^{\infty} n^{-s} = \zeta(s)$$

converges exactly for $\operatorname{Re} s > 1$.

The source's abstract pdo algebra is generated by $\mathcal{A}$, D, and $|D|$, up to smoothing remainders. In the present example these generators commute. Therefore every finite principal pdo term is a finite linear combination of operators $U^m|D|^j|D|^{-p}$, with $m \in \mathbb{Z}$ and $j,p \in \mathbb{N}_0$. For an order-zero term, $p-j \geq 0$, and

$$\operatorname{Tr}\big(U^m|D|^{j-p}|D|^{-s}\big) = \sum_{n=1}^{\infty} n^{im} n^{-(s+p-j)} = \zeta(s+p-j-im).$$

Thus all such principal terms have meromorphic continuation with at most simple poles in the discrete set

$$\{\, 1-k+im : k\in\mathbb{N}_0,\ m\in\mathbb{Z}\,\}.$$

Smoothing remainders contribute holomorphic functions after choosing the remainder order arbitrarily negative, so the triple has a simple dimension spectrum, and the minimal dimension spectrum contains the poles that actually occur.

Now take the order-zero pdo $T = U \in \mathcal{A}$. Its zeta function is

$$\zeta_{U,D}(s) = \operatorname{Tr}(U|D|^{-s}) = \sum_{n=1}^{\infty} n^{i-s} = \zeta(s-i),$$

which has a simple pole at $s = 1+i$. Hence $1+i$ belongs to the minimal dimension spectrum.

On the other hand, the D-only probe functions are

$$\zeta_{|D|^{-k},D}(s) = \operatorname{Tr}|D|^{-k}|D|^{-s} = \zeta(s+k),$$

whose only pole is $s = 1-k$, real, for $k \in \mathbb{N}_0$. Therefore

$$1+i \notin \bigcup_{k\in\mathbb{N}_0} \mathcal{P}(\zeta_{|D|^{-k},D}),$$

although $1+i$ is in the dimension spectrum. This disproves the proposed general inclusion. $\square$

Verification Notes

The construction satisfies the source's Definition 1.1 of spectral triple, Definition 1.6 of regularity, and Definition 1.22 of dimension spectrum. The only point to flag for review is scope: because $[D, \mathcal{A}] = 0$, this is a degenerate noncommutative metric geometry. That degeneracy is not excluded in the problem as stated, and the example is intentionally designed to test the unrestricted formulation.

The script `code/check_poles.py` lists the visible pole locations for the D-only probes and for the algebra element U.

Novelty and Search Bounds

Run-index search was performed for `1902.05306`, the title, and close phrases around `spectral action`, `noncommutative geometry`, `spectral triple`, and `dimension spectrum`; no direct duplicate was found. Web searches for close variants of "dimension spectrum encoded in D", "whole dimension spectrum encoded in the operator D", and "diagonal spectral triple dimension spectrum Dirichlet series" did not reveal a later explicit answer.

Limitations

This packet refutes the unrestricted general-fact formulation. It does not settle a strengthened question for nondegenerate, irreducible, geometric, or reconstruction-compatible spectral triples. In such narrower classes, the phenomenon observed by the source may still be meaningful.

Human Review Recommendation

Check that the source's $\Psi^0(\mathcal{A})$ convention indeed includes the algebra element U as an order-zero pdo. Once that is accepted, the shifted zeta pole $\zeta(s - i)$ gives the counterexample immediately.

References

[1] Michal Eckstein and Bruno Iochum, *Spectral Action in Noncommutative Geometry*, arXiv:1902.05306, 2019.

[2] Alain Connes and Henri Moscovici, *The local index formula in noncommutative geometry*, Geom. Funct. Anal. 5 (1995), 174–243.